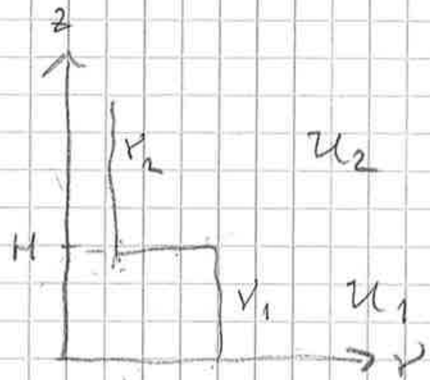
Ekmanspiral med två skikt



Ekvation:

$$\frac{d}{dz}\left(\nu \frac{d\mathcal{U}}{dz}\right) = i f \left(\mathcal{U} - \mathcal{U}_0\right) \qquad (1)$$

$$\mathcal{U} = u + i v \quad , \quad \mathcal{U}_0 = u_0 \qquad (2)$$

Randvillkor:

$$\begin{cases} \mathcal{U}_1(0) = 0 & (3) \\ \mathcal{U}_2 \to u_0 , \quad z \to \infty & (4) \end{cases}$$

Hoppvillkor:

$$\begin{cases} \mathcal{U}_1 = \mathcal{U}_2 & (5) \\ \nu_1 \frac{d\mathcal{U}_1}{dz} = \nu_2 \frac{d\mathcal{U}_2}{dz} & (6) \end{cases} \quad \text{vid } z = H.$$

Lösningen till (1) för ν = const är

$$\mathcal{U} = \text{konst} \cdot e^{\pm(1+i) z / h_E} + \mathcal{U}_0$$

där

$$h_E = \sqrt{\frac{2\nu}{f}}$$

Lösning:

$$\begin{cases} u_1 = A_1 e^{(1+i)z/h_1} + B_1 e^{-(1+i)z/h_1} + u_0 & (7) \\ u_2 = B_2 e^{-(1+i)z/h_2} + u_0 & (8) \end{cases}$$

$\Longrightarrow$

$$\begin{cases} \dfrac{\partial u_1}{\partial z} = \dfrac{1+i}{h_1}\left(A_1 e^{(1+i)z/h_1} - B_1 e^{-(1+i)z/h_1}\right) & (9) \\ \dfrac{\partial u_2}{\partial z} = -\dfrac{1+i}{h_2} B_2 e^{-(1+i)z/h_2} & (10) \end{cases}$$

där

$$h_{1,2} = \sqrt{\frac{2\nu_{1,2}}{f}} \qquad (12)$$

$(3) \Longrightarrow$

$$A_1 + B_1 + u_0 = 0 \qquad (13)$$

$(5) \Longrightarrow$

$$A_1 E_1 + \frac{B_1}{E_1} = \frac{B_2}{E_2} \qquad (14)$$

$(6) \Longrightarrow$

$$\frac{\nu_1}{h_1}(1+i)\left(A_1 E_1 - \frac{B_1}{E_1}\right) = -\frac{\nu_2}{h_2}(1+i)\frac{B_2}{E_2} \Longrightarrow$$

$$A_1 E_1 - \frac{B_1}{E_1} = -\frac{h_1}{h_2}\frac{\nu_2}{\nu_1}\frac{B_2}{E_2} = -\frac{h_2}{h_1}\frac{B_2}{E_2} \Longrightarrow$$

$$A_1 E_1 - \frac{B_1}{E_1} = -\varepsilon \frac{B_2}{E_2} \qquad (15)$$

där (16)

$$E_1 = e^{(1+i)H/h_1} \qquad (17)$$

$$E_2 = e^{(1+i)H/h_2} \qquad (18)$$

$$\varepsilon = \frac{h_2}{h_1} = \sqrt{\frac{\nu_2}{\nu_1}} \qquad (19)$$

Eliminera B_2/E_2 från (14), (15):

$$A_1 E_1 - \frac{B_1}{E_1} = -\varepsilon\left(A_1 E_1 + \frac{B_1}{E_1}\right) \Rightarrow$$

$$A_1 E_1 (1+\varepsilon) = \frac{B_1}{E_1}(1-\varepsilon) \qquad (20)$$

(13) i (20) $\Rightarrow$

$$-(B_1 + u_0) E_1^2 (1+\varepsilon) = B_1 (1-\varepsilon) \Rightarrow$$

$$B_1 \left[1 - \varepsilon + E_1^2 (1+\varepsilon)\right] = -u_0 E_1^2 (1+\varepsilon) \Rightarrow$$

$$B_1 = -u_0 \frac{E_1^2 (1+\varepsilon)}{1-\varepsilon+E_1^2(1+\varepsilon)} \qquad (21)$$

(21) och (22) $\Rightarrow$

$$A_1 = \frac{B_1}{E_1^2} \frac{1-\varepsilon}{1+\varepsilon} = -u_0 \frac{1-\varepsilon}{1-\varepsilon+E_1^2(1+\varepsilon)} \qquad (22)$$

(14) $\Rightarrow$

$$\frac{B_2}{E_2} = -u_0 \frac{E_1(1-\varepsilon) + E_1(1+\varepsilon)}{1-\varepsilon+E_1^2(1+\varepsilon)} \Rightarrow$$

$$B_2 = -u_0 \frac{2 E_1 E_2}{1-\varepsilon+E_1^2(1+\varepsilon)} \qquad (23)$$

Kontroller:

1) $h_1 = h_2 \iff E_1 = E_2 \iff \varepsilon = 1$:

$$\begin{cases} A_1 = 0 \\ B_1 = -u_0 \\ B_2 = -u_0 \end{cases}$$ Ok

2) $H \to 0 \Rightarrow E_1 = E_2 = 1 \Rightarrow$

$$\begin{cases} A_1 = -u_0 \dfrac{1-\varepsilon}{1-\varepsilon+1+\varepsilon} = -\dfrac{u_0}{2}(1-\varepsilon) \\ B_1 = -u_0 \dfrac{1+\varepsilon}{1-\varepsilon+1+\varepsilon} = -\dfrac{u_0}{2}(1+\varepsilon) \\ B_2 = -u_0 \end{cases}$$

$A_1 + B_1 + u_0 = 0$ O.K.

3) $H \to \infty \iff E_1 \to \infty \Rightarrow$

$$\begin{cases} A_1 = 0 \\ B_1 = -u_0 \\ B_2 = 0 \end{cases}$$ Ok.

Bestäm dU_1/dz vid $z=0$. (9) $\Rightarrow$

$$\left.\frac{\partial U_1}{\partial z}\right|_{z=0} = \frac{1+i}{h_1}(A_1 - B_1) \qquad (24)$$

(22), (21) $\Rightarrow$

$$A_1 - B_1 = u_0 \frac{E_1^2(1+\varepsilon) - (1-\varepsilon)}{1-\varepsilon + E_1^2(1+\varepsilon)} \tag{25}$$

Om $\varepsilon = 1$ så är $A_1 - B_1$ reellt, och vinkeln uricler $45°$. Detsamma gäller om $H/h_1 \to \infty \Rightarrow E_1^2 \to \infty$, och om $H/h_1 \to 0 \to E_1^2 = 1$.

Skriv om (25):

$$A_1 - B_1 = u_0 \frac{E_1^2 - \frac{1-\varepsilon}{1+\varepsilon}}{E_1^2 + \frac{1-\varepsilon}{1+\varepsilon}} = u_0 \frac{E_1^2 - R}{E_1^2 + R} \tag{26}$$

där

$$R = \frac{1-\varepsilon}{1+\varepsilon} \tag{27}$$

Sätt

$$\varphi = \frac{H}{h_1} \tag{28}$$

Antag $\varphi \ll 1$. $\Rightarrow$

$$E_1 = e^{(1+i)\varphi} \simeq 1 + (1+i)\varphi + \tfrac{1}{2}(1+i)^2\varphi^2 \tag{29}$$

(29) i (26) $\Rightarrow$

$$A_1 - B_1 \simeq u_0 \frac{1 + 2(1+i)\varphi - R}{1 + 2(1+i)\varphi + R} =$$

$$= u_0 \frac{1-R}{1+R} \frac{1 + 2\frac{1+i}{1-R}\varphi}{1 + 2\frac{1+i}{1+R}\varphi} \simeq u_0 \frac{1-R}{1+R}\left[1 + 2(1+i)\varphi\left(\frac{1}{1-R} - \frac{1}{1+R}\right)\right]$$

Vi har

$$1 - R = 1 - \frac{1-\varepsilon}{1+\varepsilon} = \frac{2\varepsilon}{1+\varepsilon}$$

$$1+R = 1+\frac{1-\varepsilon}{1+\varepsilon} = \frac{2}{1+\varepsilon}$$

$$\frac{1}{1-R} - \frac{1}{1+R} = \frac{1+\varepsilon}{2\varepsilon} - \frac{1+\varepsilon}{2} = \frac{1+\varepsilon}{2}\left(\frac{1}{\varepsilon}-1\right) = \frac{1-\varepsilon^2}{2\varepsilon}$$

$$\frac{1-R}{1+R} = \varepsilon$$

$$A_1 - B_1 \simeq u_0\,\varepsilon\left[1 + 2\varphi(1+i)\,\frac{1-\varepsilon^2}{2\varepsilon}\right] \qquad (30)$$

Antag att $\varepsilon \ll 1$, och $\varphi \ll 1$, men att vi inte känner ε/φ. $(27) \Rightarrow$

$$R \simeq 1 - 2\varepsilon \qquad (31)$$

(29) och (31) i (26) $\Rightarrow$

$$A_1 - B_1 \simeq u_0\,\frac{1+2(1+i)\varphi - 1 + 2\varepsilon}{1+2(1+i)\varphi + 1 - 2\varepsilon} \simeq$$

$$\simeq u_0\left[(1+i)\varphi + \varepsilon\right] \Rightarrow \qquad (32)$$

$$\left.\frac{\partial u_1}{\partial z}\right|_{z=0} = u_0\,\frac{1+i}{h_1}\left[(1+i)\varphi + \varepsilon\right] \qquad (33)$$

Vi ser att vinkeln vrider mer än 45°.

Bestäm $u(H)$. $(7) \Rightarrow$

$$u(H) = u_1(H) = A_1E_1 + \frac{B_1}{E_1} + u_0 = \left[(21),(22)\right] =$$

$$= -u_0\,\frac{E_1(1-\varepsilon) + E_1(1+\varepsilon)}{1-\varepsilon + E_1^2(1+\varepsilon)} + u_0 =$$

$$= u_0\left(1 - \frac{2E_1}{1-\varepsilon+E_1^2(1+\varepsilon)}\right) =$$

$$= u_0\,\frac{1-\varepsilon+E_1^2+\varepsilon E_1^2 - 2E_1}{1-\varepsilon+E_1^2(1+\varepsilon)} \Rightarrow$$

$$\mathcal{U}(H) = u_0 \frac{\varepsilon(E_1^2-1) + (E_1-1)^2}{E_1^2+1+\varepsilon(E_1^2-1)} =$$

$$= u_0(E_1-1)\frac{\varepsilon(E_1+1) + E_1-1}{E_1^2+1+\varepsilon(E_1^2-1)} \qquad (33)$$

För $\varepsilon \ll 1$ och $\varphi \ll 1$ får vi

$$\mathcal{U}(H) \simeq u_0(1+i)\varphi \frac{(1+i)\varphi + 2\varepsilon}{2} =$$

$$= u_0\left(i\varphi^2 + (1+i)\varepsilon\varphi\right) \qquad (34)$$

Kontrollera att (34) stämmer med en Taylorutveckling av $\mathcal{U}(z)$ baserad på (33). Vi har

$$\mathcal{U}(H) \simeq \left.\frac{d\mathcal{U}}{dz}\right|_{z=0} H + \frac{d^2\mathcal{U}}{dz^2}\frac{H^2}{2} + \ldots \qquad (35)$$

(1) $\Rightarrow$

$$\left.\frac{d^2\mathcal{U}}{dz^2}\right|_{z=0} = i\frac{f}{Y_1}(-u_0) = -i\,u_0\,\frac{2}{h_2^2} \qquad (36)$$

(33) och (36) i (35) $\Rightarrow$

$$\mathcal{U}(H) \simeq u_0(1+i)\varphi\left[(1+i)\varphi + \varepsilon\right] - i u_0\varphi^2 =$$

$$= u_0\left\{\left[(1+i)^2 - i\right]\varphi^2 + (1+i)\varepsilon\varphi\right\} =$$

$$= u_0\left(i\varphi^2 + (1+i)\varepsilon\varphi\right) =$$

$$= u_0(1+i)\varphi\left[\tfrac{1}{2}(1+i)\varphi + \varepsilon\right] \qquad (37)$$

OK